

Micro-Lending & Peer Credit Risk Assessor

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose: This document specifies the requirements of the Micro-Lending & Peer Credit Risk Assessor, a system that lets borrowers submit financial details, assigns a credit risk rating, and lets peer lenders view that rating before lending.

1.2 Scope: The system covers borrower profile submission, credit risk computation, risk warnings, lender viewing of risk ratings, loan approval, and repayment schedule generation, with secure and consistent handling of financial data.

1.3 Definitions: EMI - Equated Monthly Instalment. DTI - Debt-to-Income ratio. ACID - Atomicity, Consistency, Isolation, Durability. FR - Functional Requirement. NFR - Non-Functional Requirement.

2. Overall Description

2.1 Product Perspective: A web-based application with a borrower interface, a lender interface, an application layer, and a database.

2.2 User Classes: Borrower - submits financial information and applies for a loan. Peer Lender - views risk rating and approves loans. Administrator - manages users and system data.

2.3 Operating Environment: Standard web browser, application server, and relational database.

2.4 Assumptions and Dependencies: Borrowers provide accurate financial information; the database supports ACID transactions.

3. System Features (Functional Requirements)

ID	Description	Priority
FR-001	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High
FR-002	The system shall allow a	High

	borrower to submit the financial information required for credit risk assessment.	
FR-003	The system shall allow a peer lender to view the borrower's credit risk rating before lending.	High
FR-004	The system shall generate a structured repayment schedule for an approved micro-loan.	High
FR-005	The system shall trigger an automated warning when a borrower's credit risk is classified as High.	High

4. Non-Functional Requirements

ID	Description	Priority
NFR-001	The system shall execute repayment transactions and EMI distribution inside database ACID transactions to prevent double-spending.	High
NFR-002	The system shall protect borrower financial information from unauthorized access.	High

5. External Interface Requirements

- User Interface: web forms for borrower financial details; lender view showing the Low/Medium/High risk tier.
- Software Interface: relational database for storing users, profiles, loans, schedules and transactions.
- Hardware Interface: none specific; runs on standard server and client devices.

6. Constraints

- Repayment and EMI distribution must run inside database ACID transactions.
- Borrower financial data must be accessible only to authorized users.

7. Acceptance Criteria

Acceptance criteria for each requirement are listed in FR_and_NFR.docx (folder 1-RE) and traced to test cases in RTM_Table.docx.